

Physics Lecture 2: Electromagnetism

Introduction to Electric Fields and Forces

Electromagnetism is the branch of physics concerning the study of the electromagnetic force, which is a type of physical interaction that occurs between electrically charged particles. This force is the foundation of electricity and magnetism, as well as the behavior of light.

Key Concepts and Formulas

- **Coulomb's Law:** Describes the electrostatic force between two point charges. $F = k r^2 |q_1 q_2|$
 - F = electrostatic force
 - k = Coulomb's constant, $8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$
 - q_1, q_2 = magnitudes of the two charges
 - r = distance between the charges
- **Electric Field:** A vector field that surrounds an electric charge and exerts a force on other charges. $E = q_0 F$
For a point charge, the magnitude of the electric field is: $E = k r^2 |q|$
- **Ohm's Law:** Relates voltage, current, and resistance in an electrical circuit. $V = IR$
 - V = voltage (in Volts, V)
 - I = current (in Amperes, A)
 - R = resistance (in Ohms, Ω)

Mathematical Example Problem: Electric Field

Problem: Two point charges, $q_1 = +2.0 \text{ nC}$ and $q_2 = -3.0 \text{ nC}$, are separated by a distance of 0.05 m . Find the magnitude of the electric field at a point P located exactly halfway between them.

Solution:

1. **Identify the contributions from each charge:** The electric field at point P is the vector sum of the fields produced by each charge. The point P is $r_1 = r_2 = 0.025 \text{ m}$ from each charge.
2. **Calculate the electric field from q_1 (E_1):** Since q_1 is positive, its electric field points away from it, towards q_2 . $E_1 = k r^2 |q_1| = (8.99 \times 10^9)(0.025)^2 |2.0 \times 10^{-9}|$ $E_1 = 28784 \text{ N/C}$
3. **Calculate the electric field from q_2 (E_2):** Since q_2 is negative, its electric field points towards it, also in the same direction as E_1 . $E_2 = k r^2 |q_2| = (8.99 \times 10^9)(0.025)^2 |-3.0 \times 10^{-9}|$ $E_2 = 43152 \text{ N/C}$
4. **Find the net electric field (E_{net}):** Since both fields are in the same direction, we simply add their magnitudes. $E_{\text{net}} = E_1 + E_2 = 28784 + 43152 = 71936 \text{ N/C}$
The net electric field at point P is 71936 N/C and points from q_1 towards q_2 .